TAMU-CC · MELARAM LAB

Cotton Field Log — the complete guide

Hi Emma. This replaces the spreadsheet, the R scripts and the macro button. There is nothing to install and nothing to code. You open a link, type numbers, and the Excel file is one click away whenever you or a mentor wants it.

Everything below is shown with the built-in sample trial. Load it yourself — **Setup → Load sample data** — and click around freely. Clearing it puts your real data back exactly as it was, so there is nothing you can break by exploring.

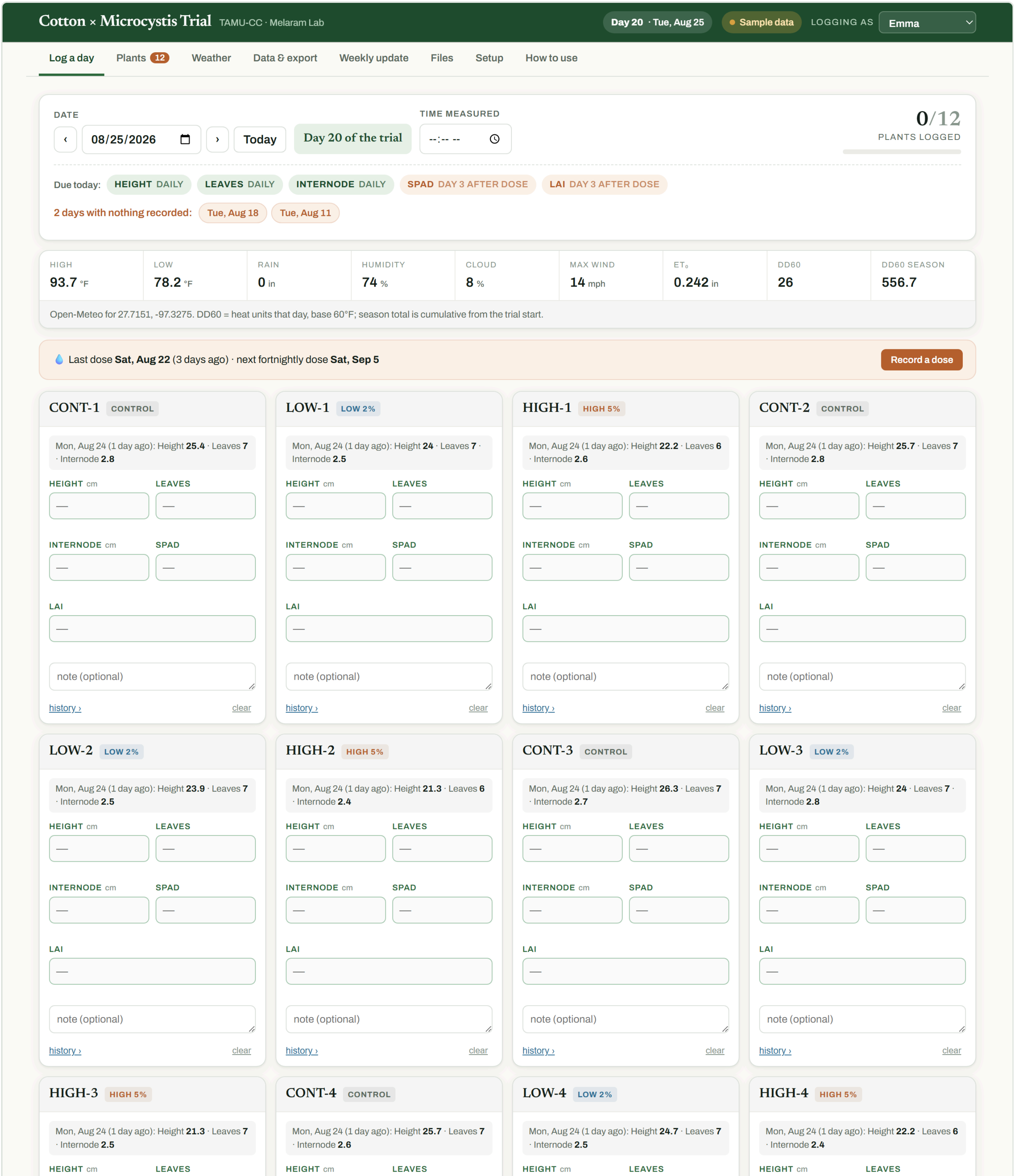
Cotton × *Microcystis* trial · Emma Doria
Live app: **aidanjmeyers.github.io/ed-cotton-growth-dashboard**
Generated August 25, 2026

The 60-second version

1. Open the link. Check **Logging as** (top right) says your name.
2. Type each plant's numbers on the **Log a day** tab. It saves itself.
3. On dosing days, press **Record a dose**.
4. When someone wants the data: **Data & export** → **Download Excel**.
5. When someone wants an update: **Status update** → **Copy to clipboard**.

Everything else in this guide is detail you can come back to.

1. Reading the top bar

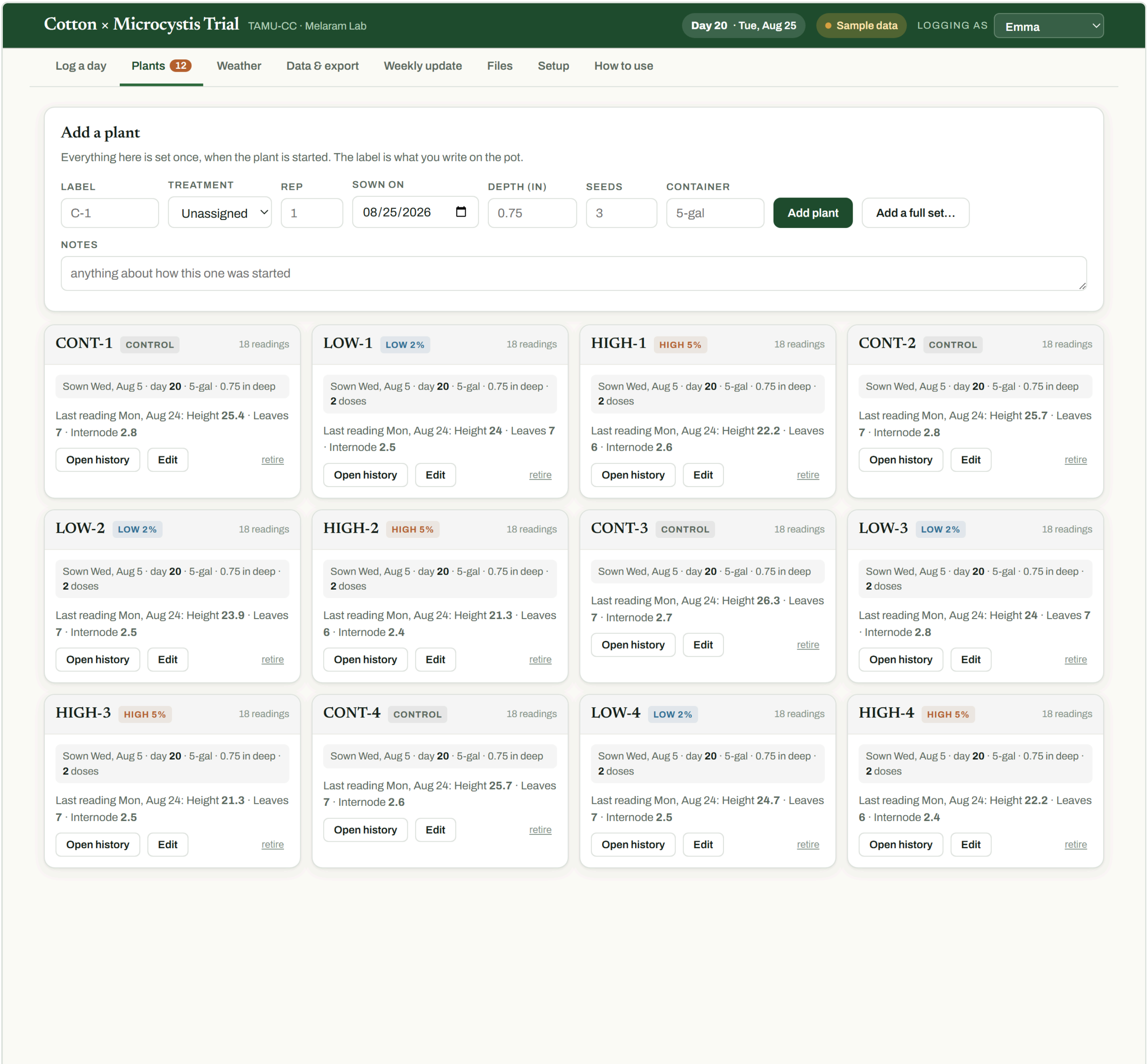


Day 20 · Tue, Aug 25	How many days into the trial today is
Saved & synced	Everything you typed has reached the database. See section 12 for the other things this can say
Logging as	Your name goes onto every reading you record. Check it before you start

The tabs across the top are the eight places anything happens. They are covered in order below.

2. First time only — add your plants

Go to **Plants**.



Add one plant with the label you wrote on the pot, its treatment group, when it was sown, how deep, and how many seeds.

Add a full set is faster when they all start together: fill in the shared details, type a prefix in **Label**, press **Add a full set…**, and say how many. It creates numbered plants split evenly across Control, Low 2% and High 5%.

Each plant tile then shows its age, container, sowing depth, how many doses it has had, its total readings, and its last reading.

- **Edit** — rename, change treatment, change notes
- **Retire** — for a plant that dies or leaves the trial. Its history is kept in full, it just stops appearing on the daily list. You can un-retire it

Adding a plant mid-trial is fine. So is retiring one.

3. The daily routine

Everything on the **Log a day** tab, top to bottom.

The date row

The date is already today. The arrows step one day at a time, **Today** jumps back. Backfilling an earlier day is completely normal — just pick the date.

Time measured is when you took the readings, not when you typed them.

12/12 plants logged tracks the day's progress.

Due today

The green band says what the schedule expects:

MEASUREMENT	WHEN
Height, leaves, internode	Every day
SPAD	Weekly, Fridays
Ceptometer (LAI)	Every other Friday
SPAD and LAI again	3 days after any dose

You never have to work the post-dose timing out yourself — the band says `DAY 3 AFTER DOSE` in orange when it applies. Boxes that are due are outlined in green.

Days with nothing recorded

Orange chips list days you missed. Tap one to jump straight there and fill it in.

The weather strip

Filled in for you from the plot's coordinates — you never fetch this.

DD60 is the heat units for that day; **DD60 SEASON** is the running total since the trial started. Cotton development tracks heat units better than calendar days, which is why it is there.

The plant cards

The grey line on each card is what that plant looked like last time, so you always know what you are comparing against.

Type the numbers. **There is no save button** — about a second after you stop typing the card turns green and says ✓ saved.

Under each box is a small arrow comparing what you just typed against that plant's last reading of the same thing: green up, red down. A red arrow is not an error. It is the plant telling you something.

Three time-savers:

- **Enter** jumps to the same box on the next plant, so you can go down the bench without touching the mouse
- **+ also record** reveals measurements that are not due today, if you took them anyway
- **history ›** opens that plant's full record

Two rules that matter for the analysis

Blank is not zero. An empty box means you did not measure it. A zero means you measured and it was zero. Statistics treat those completely differently, so please leave things blank rather than typing 0.

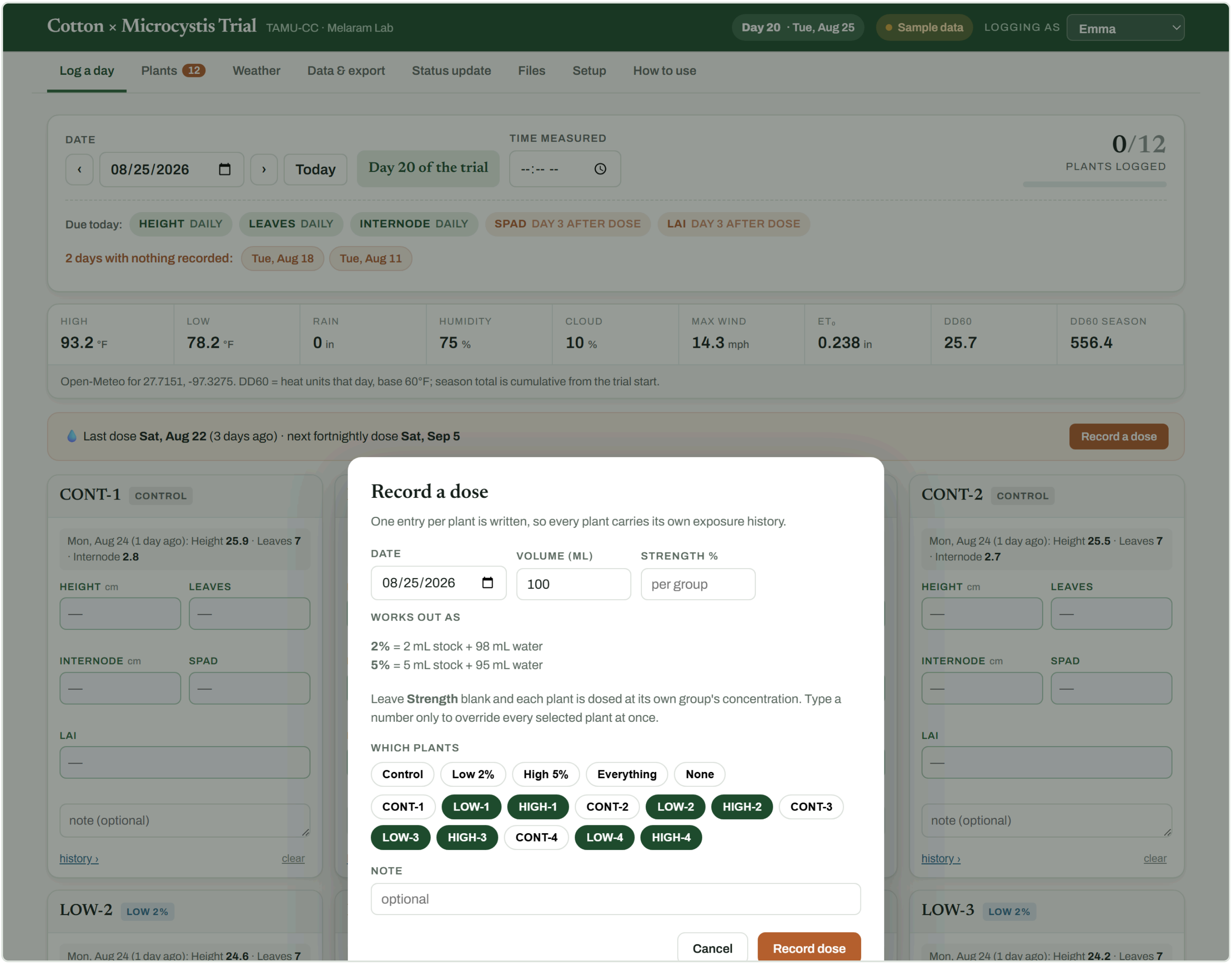
Leaves is a total count — how many are on the plant right now, not how many are new. If it drops from 6 to 4 because two fell off, put 4. That fall is exactly the signal this trial is looking for.

The day note

At the bottom: one note for the whole day. Watering, dosing, weather you saw, anything odd. This is where "the AC in the greenhouse failed overnight" belongs.

4. Dosing days

Press **Record a dose** in the orange strip.



Pick the date, then the plants — **clicking a group name selects that whole group**. Controls are left out by default.

Leave **Strength** blank and each plant is dosed at its own group's concentration. The panel spells out the actual mixture so you can make it up correctly:

```
2% = 2 mL stock + 98 mL water
5% = 5 mL stock + 95 mL water
```

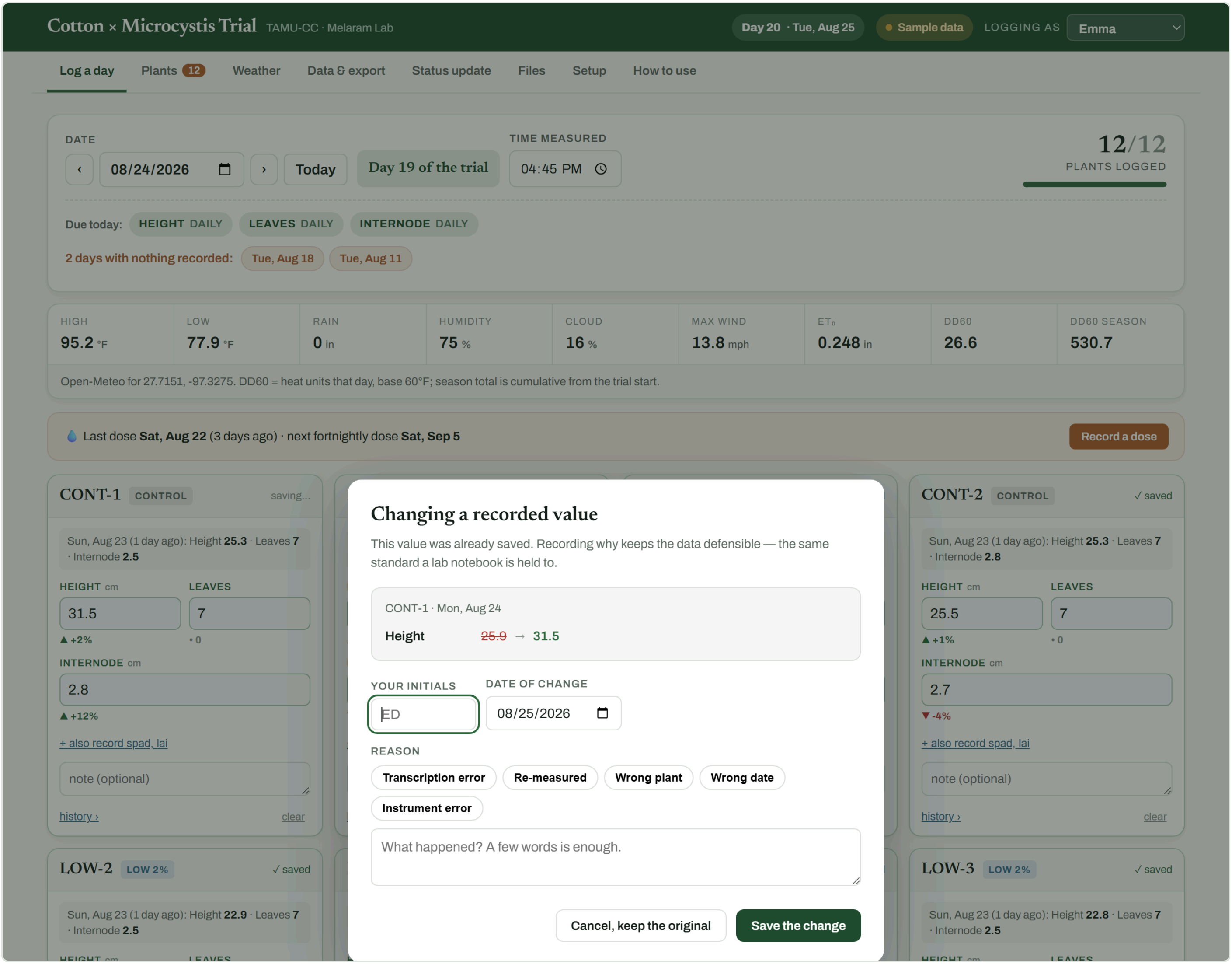
Type a number in **Strength** only if you want to override every selected plant at once.

One record is written per plant, so each plant carries its own exposure history. From then on the app knows when the 3-day post-dose readings are due, every chart marks the dose day with a dashed line, and the strip tells you when the next fortnightly dose falls.

5. Correcting something already saved

Type over a value, or press **clear** to remove a reading entirely.

If that value had already been committed — you saved it on an earlier visit, or someone else recorded it — this appears first:



Your initials	Remembered after the first time
Date of change	Defaults to today
Reason	Tap a common one, or type your own

It shows exactly what is about to change, and **nothing is saved until it is filled in**. **Cancel, keep the original** puts the value back.

Correcting something you typed a minute ago does *not* ask — that is just finishing your entry. It only asks once a reading has settled, and once a reading has been formally corrected it asks every time after that.

Why it matters: every number in your dataset either came straight off the instrument, or has your initials and a reason next to it explaining why it differs. If a committee asks "why is this point different from your raw sheet", the answer is in the file. The original value is never deleted.

Corrections appear on the plant's own page, on the **Corrections** sheet of the Excel export, and in the permanent record.

6. One plant at a time

Click any plant's name, anywhere.

Cotton × Microcystis Trial TAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25 · Sample data · LOGGING AS Emma

Log a day · Plants 12 · Weather · Data & export · Weekly update · Files · Setup · How to use

← All plants

HIGH-1 HIGH 5%

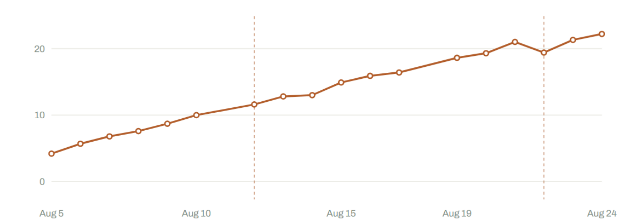
Sown **Wed, Aug 5** · **20** days old Container **5-gal** Depth **0.75 in** Seeds **3** Rep **1** Readings **18** Doses **2**

Edit plant Log a reading today

HEIGHT	LEAVES	INTERNODE	SPAD	LAI
22.2 cm	6	2.6 cm	40.4	1.1
▲+4%	• 0	▲+4%	▲+2%	▲+39%
Aug 24	Aug 24	Aug 24	Aug 21	Aug 21

Height over time

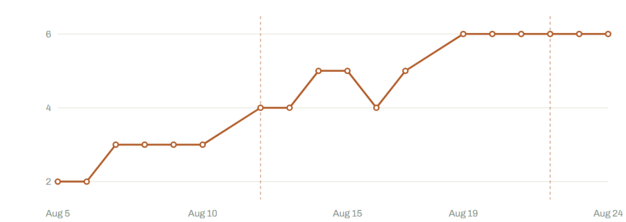
cm · dashed lines are dose days



dose day

Leaves over time

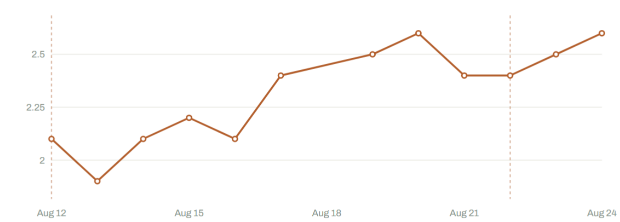
count · dashed lines are dose days



dose day

Internode over time

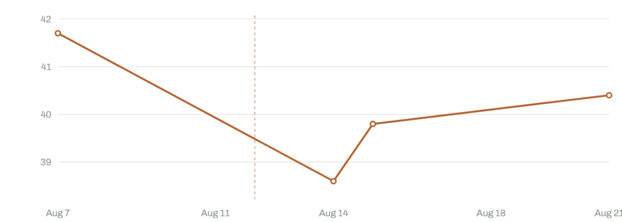
cm · dashed lines are dose days



dose day

SPAD over time

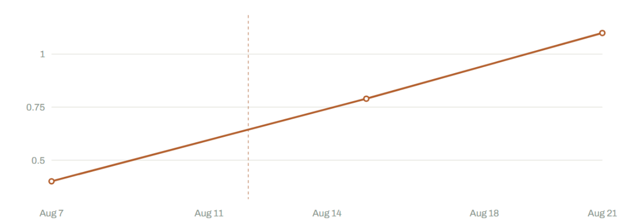
SPAD value · dashed lines are dose days



dose day

LAI over time

leaf area index · dashed lines are dose days



dose day

Every reading

Arrows compare each value with this plant's previous reading of the same thing. Dose days are marked.

DATE	DAP	HEIGHT	LEAVES	INTERNODE	SPAD	LAI	DOSE	NOTE
2026-08-24	19	22.2 ▲+4%	6	2.6 ▲+4%	—	—	—	
2026-08-23	18	21.3 ▲+10%	6	2.5 ▲+4%	—	—	—	
2026-08-22	17	19.4 ▼-8%	6	2.4	—	—	5% · 100 ML	
2026-08-21	16	21 ▲+9%	6	2.4 ▼-8%	40.4 ▲+2%	1.1 ▲+39%	—	
2026-08-20	15	[illegible]	6	[illegible]				

2026-08-20	15	19.3 ▲+4%	6	2.6 ▲+4%	—	—	—
2026-08-19	14	18.6 ▲+13%	6 ▲+1	2.5 ▲+4%	—	—	—

You get its sowing details, its latest value for everything with the change since the reading before, a growth curve per measurement with dose days marked, every reading it has ever had with up/down arrows, and — if there are any — its corrections.

This is the screen to open when someone asks "how is the high-dose group doing", or when you want to see whether a plant turned the corner after an exposure.

7. Weather

Cotton × Microcystis Trial TAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25 | Sample data | LOGGING AS Emma

Log a day | Plants 12 | **Weather** | Data & export | Weekly update | Files | Setup | How to use

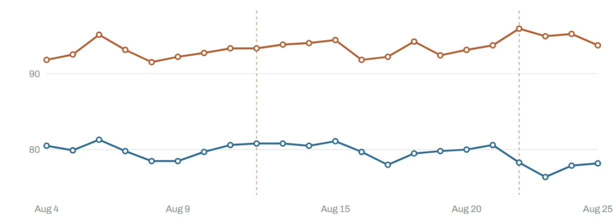
DAYS OF WEATHER	MEAN HIGH	MEAN LOW	HOTTEST	TOTAL RAIN	HEAT UNITS DD60
22	93.4 °F	79.6 °F	95.9 °F	0 in	556.7

FROM mm/dd/yyyy TO mm/dd/yyyy

Refresh weather | ⇩ Weather to Excel | ⇩ CSV

Temperature

daily high and low at the plot, °F



High · Low · dose day

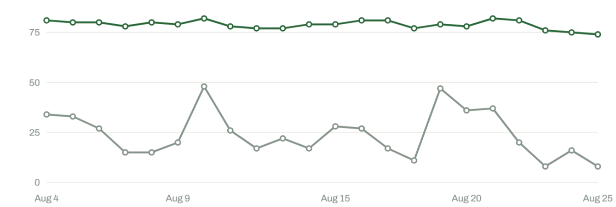
Rainfall

inches per day

No rain at the plot in this period.

Humidity and cloud

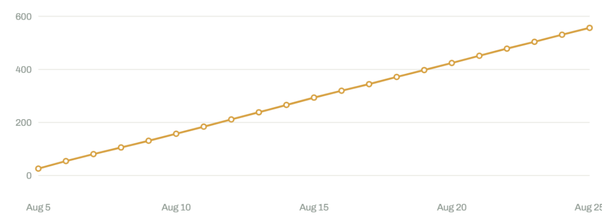
daily means, %



Humidity · Cloud

Cumulative heat units

DD60 since the trial start — the cotton growth clock

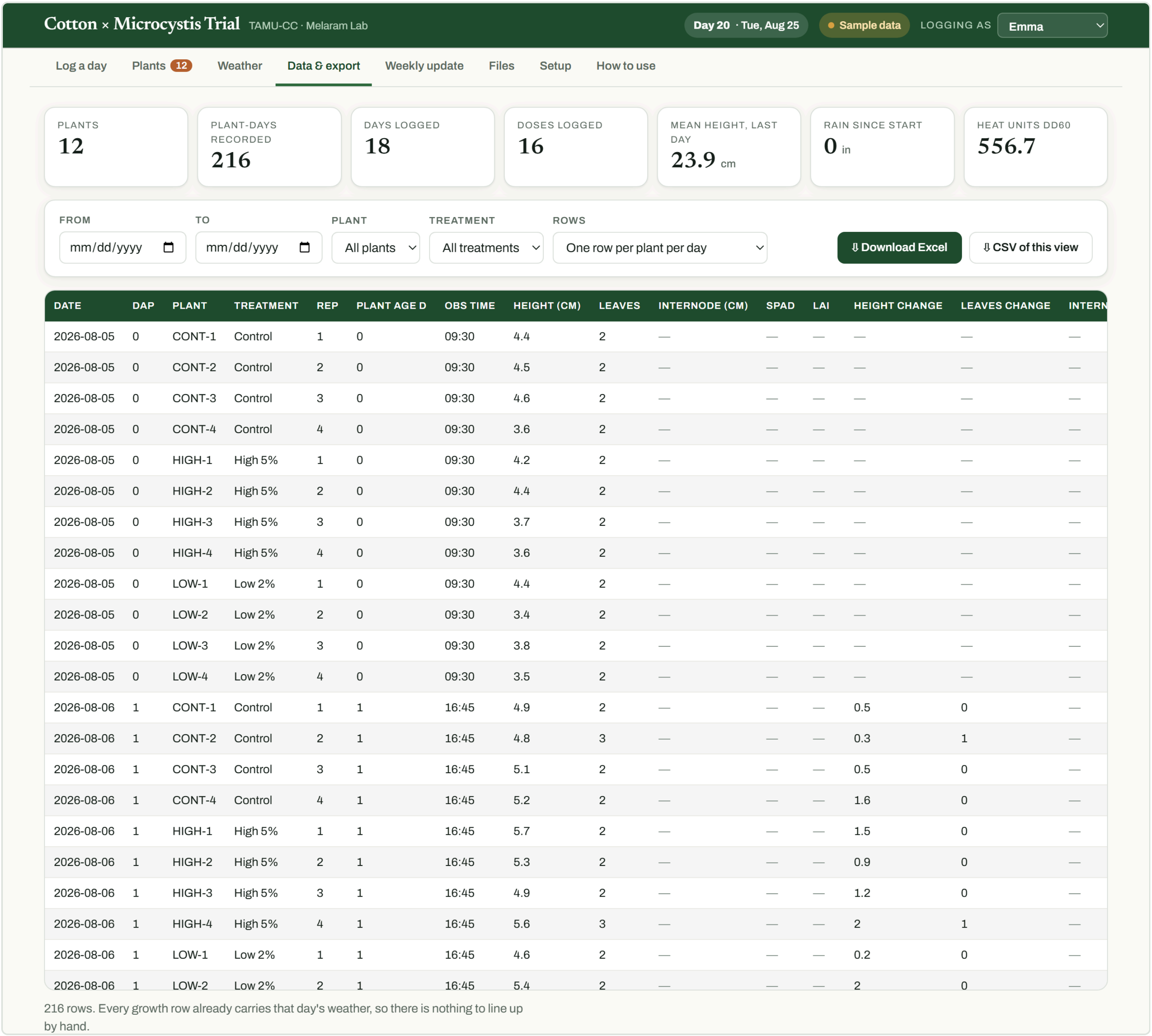


DATE	DAP	TEMP HIGH F	TEMP LOW F	TEMP MEAN F	RAIN IN	HUMIDITY %	CLOUD %	WIND MAX MPH	ETO IN	SOLAR MJ/M2	DD60	DD60 CUMULATIVE	SOURCE
2026-08-25	20	93.7	78.2	85.4	0	74	8	14	0.242	26.1	26	556.7	open-meteo
2026-08-24	19	95.2	77.9	85.7	0	75	16	13.8	0.248	26.6	26.6	530.7	open-meteo
2026-08-23	18	94.9	76.4	85.1	0	76	8	11.1	0.241	26.4	25.7	504.1	open-meteo
2026-08-22	17	95.9	78.3	85.5	0	81	20	13.3	0.233	26.3	27.1	478.4	open-meteo
2026-08-21	16	93.7	80.6	85.7	0	82	37	15.5	0.226	26	27.2	451.3	open-meteo
2026-08-20	15	93.1	80	86.4	0	78	36	17.6	0.24	26.5	26.6	424.1	open-meteo
2026-08-19	14	92.4	79.8	85.4	0	79	47	15.4	0.235	26.4	26.1	397.5	open-meteo
2026-08-18	13	94.2	79.5	85.7	0	77	11	15.5	0.238	25.9	26.8	371.4	open-meteo
2026-08-17	12	92.2	78	85.3	0	81	17	13.5	0.223	25.8	25.1	344.6	open-meteo
2026-08-16	11	91.8	79.7	85.4	0	81	27	12.7	0.208	23.9	25.8	319.5	open-meteo
2026-08-15	10	94.4	81.1	86.3	0	79	28	15	0.223	24.2	27.8	293.7	open-meteo
2026-08-14	9	94	80.5	86	0	79	17	18.3	0.215	22.5	27.3	265.9	open-meteo
2026-08-13	8	93.8	80.8	86.6	0	77	22	19.1	0.252	26.7	27.3	238.6	open-meteo
2026-08-12	7	93.3	80.8	86.2	0	77	17	20.1	0.242	25.7	27.1	211.3	open-meteo
2026-08-11	6	93.3	80.6	86.8	0	78	26	16.6	0.234	25.2	26.9	184.2	open-meteo

Conditions at the plot for the whole trial: temperature, rainfall, humidity and cloud, and cumulative heat units. Filter by date, and export just the weather to Excel or CSV.

You never have to fetch this — it updates itself. **Refresh weather** only forces it. New days appear about a day late, which is normal for reanalysis data.

8. Data & export



Cotton × Microcystis Trial TAMU-CC · Melaram Lab

Day 20 · Tue, Aug 25 · Sample data · LOGGING AS Emma

Log a day | Plants 12 | Weather | Data & export | Weekly update | Files | Setup | How to use

PLANTS	PLANT-DAYS RECORDED	DAYS LOGGED	DOSES LOGGED	MEAN HEIGHT, LAST DAY	RAIN SINCE START	HEAT UNITS DD60
12	216	18	16	23.9 cm	0 in	556.7

FROM mm/dd/yyyy · TO mm/dd/yyyy · PLANT All plants · TREATMENT All treatments · ROWS One row per plant per day · ⇩ Download Excel · ⇩ CSV of this view

DATE	DAP	PLANT	TREATMENT	REP	PLANT AGE D	OBS TIME	HEIGHT (CM)	LEAVES	INTERNODE (CM)	SPAD	LAI	HEIGHT CHANGE	LEAVES CHANGE	INTERN
2026-08-05	0	CONT-1	Control	1	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	CONT-2	Control	2	0	09:30	4.5	2	—	—	—	—	—	—
2026-08-05	0	CONT-3	Control	3	0	09:30	4.6	2	—	—	—	—	—	—
2026-08-05	0	CONT-4	Control	4	0	09:30	3.6	2	—	—	—	—	—	—
2026-08-05	0	HIGH-1	High 5%	1	0	09:30	4.2	2	—	—	—	—	—	—
2026-08-05	0	HIGH-2	High 5%	2	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	HIGH-3	High 5%	3	0	09:30	3.7	2	—	—	—	—	—	—
2026-08-05	0	HIGH-4	High 5%	4	0	09:30	3.6	2	—	—	—	—	—	—
2026-08-05	0	LOW-1	Low 2%	1	0	09:30	4.4	2	—	—	—	—	—	—
2026-08-05	0	LOW-2	Low 2%	2	0	09:30	3.4	2	—	—	—	—	—	—
2026-08-05	0	LOW-3	Low 2%	3	0	09:30	3.8	2	—	—	—	—	—	—
2026-08-05	0	LOW-4	Low 2%	4	0	09:30	3.5	2	—	—	—	—	—	—
2026-08-06	1	CONT-1	Control	1	1	16:45	4.9	2	—	—	—	0.5	0	—
2026-08-06	1	CONT-2	Control	2	1	16:45	4.8	3	—	—	—	0.3	1	—
2026-08-06	1	CONT-3	Control	3	1	16:45	5.1	2	—	—	—	0.5	0	—
2026-08-06	1	CONT-4	Control	4	1	16:45	5.2	2	—	—	—	1.6	0	—
2026-08-06	1	HIGH-1	High 5%	1	1	16:45	5.7	2	—	—	—	1.5	0	—
2026-08-06	1	HIGH-2	High 5%	2	1	16:45	5.3	2	—	—	—	0.9	0	—
2026-08-06	1	HIGH-3	High 5%	3	1	16:45	4.9	2	—	—	—	1.2	0	—
2026-08-06	1	HIGH-4	High 5%	4	1	16:45	5.6	3	—	—	—	2	1	—
2026-08-06	1	LOW-1	Low 2%	1	1	16:45	4.6	2	—	—	—	0.2	0	—
2026-08-06	1	LOW-2	Low 2%	2	1	16:45	5.4	2	—	—	—	2	0	—

216 rows. Every growth row already carries that day's weather, so there is nothing to line up by hand.

The **Rows** dropdown switches what you are looking at:

VIEW	ONE ROW PER
Plant per day	Every individual reading — the raw data
Day	Whole-trial summary per day: means, standard deviations, counts
Treatment per day	Each group's mean and sd per day — the view to run stats from
Dosing log	Every exposure

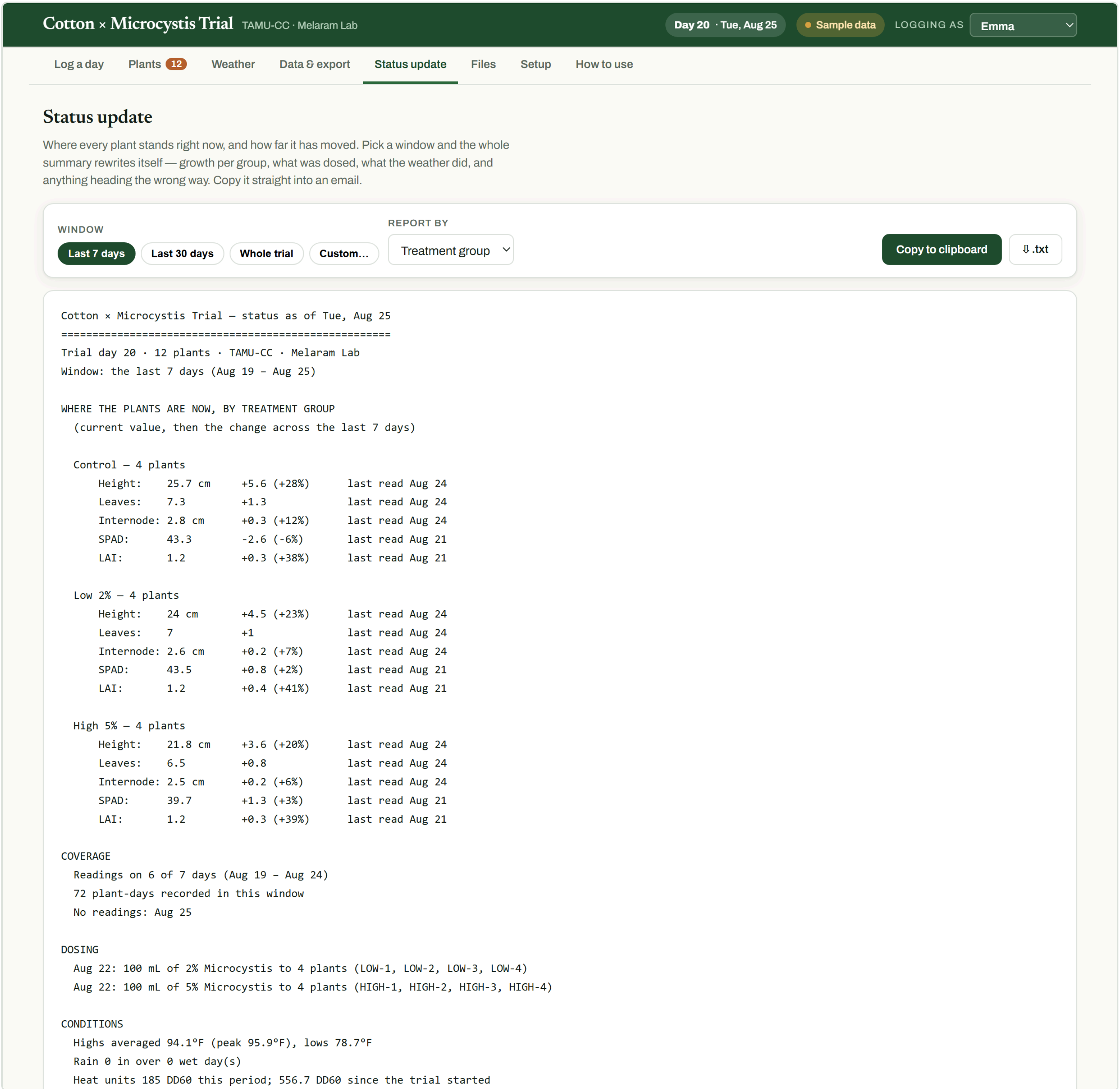
Narrow it by date range, single plant, or treatment group. **CSV of this view** downloads exactly what is on screen.

Download Excel gives you one workbook with everything:

SHEET	WHAT'S ON IT
Growth	One row per plant per day, with change columns and that day's weather attached
Daily	Whole-trial summary per day
Treatment means	Per group per day — mean, sd, n
Doses	Every exposure: which plant, what strength, how much
Corrections	Every value changed after it was recorded, with initials and reason
Weather	Daily conditions and heat units
Plants	Your plant list with treatments and sowing details
Data dictionary	What every column means, in plain English

Because the weather is already on every growth row, there is no lining-up by hand and no VLOOKUP to write.

9. Status update



Pick a window — **last 7 days**, **last 30 days**, **whole trial**, or custom — and the summary rewrites instantly:

```
High 5% — 4 plants
    Height:    21.8 cm      +3.6 (+20%)
    Leaves:    6.5          +0.8
    SPAD:      39.7         +1.3 (+3%)       last read Aug 21
```

Then coverage, dosing, the weather, anything heading the wrong way, and your field notes. **Copy to clipboard** and paste it into an email.

A measurement that is not taken daily still shows its current value with the date it was last read, so a 7-day report is not full of blanks. And a 30-day window on a 20-day trial says so rather than pretending.

Report by switches between treatment groups and every plant individually.

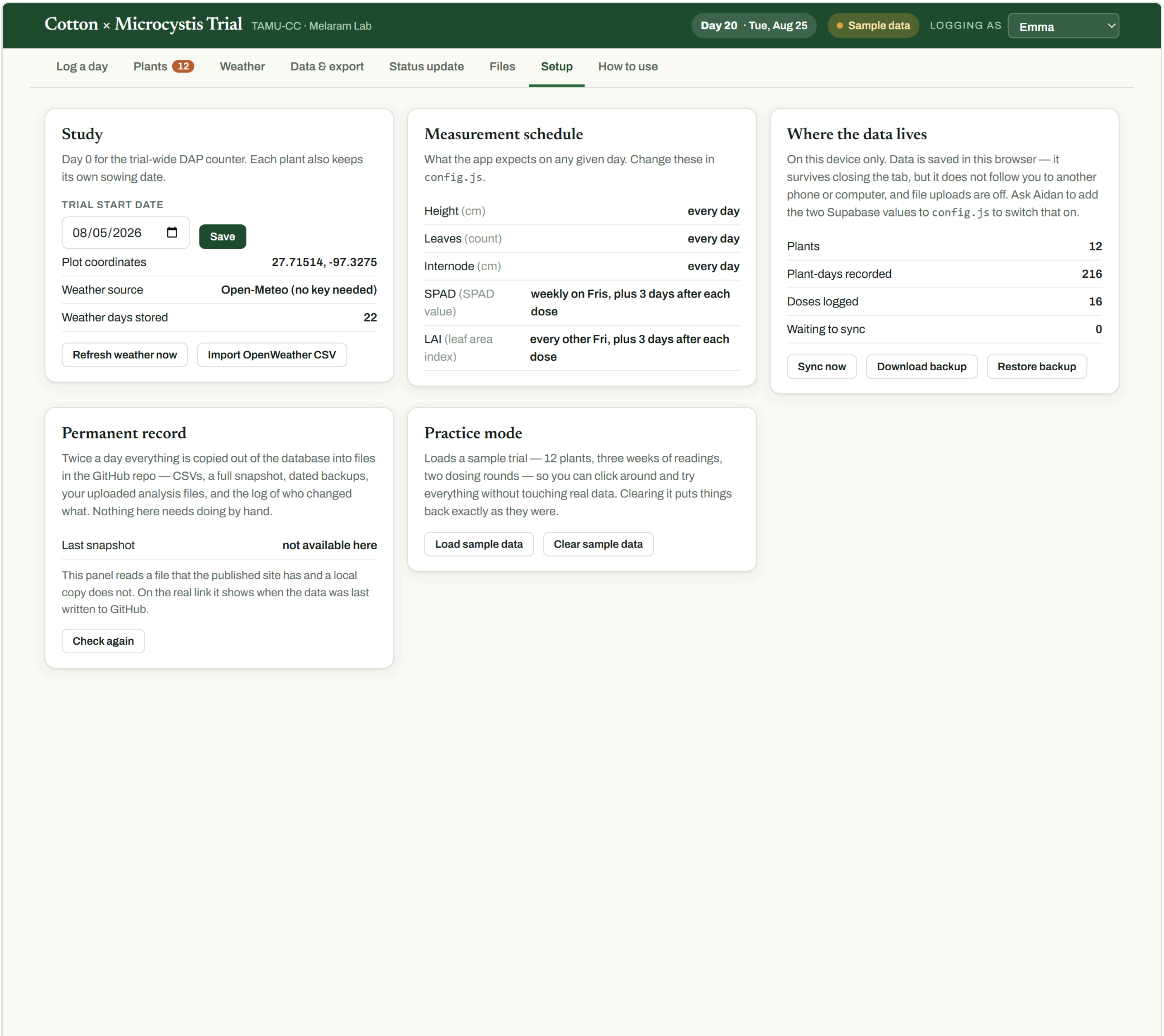
10. Files — your working analysis

Drop your analysis workbook here whenever you update it. Anyone with the link can then see the live data *and* download your latest analysis, without you emailing anything. Protocols and photos are welcome too.

Label each upload (working file / protocol / photo / other) and add a note like "week 3, updated ANOVA".

Nothing is ever overwritten. Every upload is kept and tagged with your name and date, and the newest working file is marked `latest` so nobody grabs a stale one. All of them are copied into the permanent record twice a day.

11. Setup



Four panels. You will rarely need any of them, but here is what they do.

Study — the trial start date, the plot coordinates, and buttons to refresh the weather or import a weather file from the old R pipeline.

Measurement schedule — what the app expects and when. Changing it is a config edit, so ask Aidan.

Where the data lives — how many plants, readings and doses exist, and how many changes are waiting to sync. Also:

- **Sync now** — push anything waiting, when you have signal

- **Download backup** — saves the entire season to one file you can email
- **Restore backup** — loads one back in

Permanent record — when your data was last written to permanent files, how many changes have been logged, how many backups are kept, and who has been active this week. If that line turns red, the automatic backup has stopped — tell Aidan. It does not mean anything is lost.

Practice mode — loads a full sample trial to click around in. Clearing it restores your real data untouched.

12. If something looks wrong

The pill at the top right tells you the state of your data:

IT SAYS	MEANING
Saved & synced	Everything has reached the database. Normal
Offline — saved here	No signal. Your typing is safe on this device and will sync automatically
N waiting	That many changes still to send. Open the app on wifi, or press **Sync now**
On this device	Not connected to the shared database at all — tell Aidan
Sample data	You are in practice mode. Nothing you type is real

It works without signal. Anything typed in the greenhouse is held on the phone and syncs the moment you have bars again. Just do not clear your browser data while something is waiting.

Nothing is deleted by accident. The only removal is the `clear` link on a single card, and it asks for a reason first.

If a number looks wrong, there is a record of what it was before and who changed it.

If you are stuck, press **Setup** → **Download backup**, email that file to Aidan, and say what looked wrong. You cannot break this by clicking around.

13. Where your data actually lives

You do not have to do anything for this, but it is worth knowing.

Everything you type goes into a shared database immediately, which is why it appears on every device. Twice a day, all of it is also written into permanent files on GitHub: spreadsheets of every reading, dated backups, your uploaded analysis files, and a log of every change with who made it and when.

Practically: you cannot lose this data by dropping your phone.

The one habit that matters

Log every day you visit, even if it is a short visit and even if half the boxes are blank. Consistency is what makes the trends real. The orange "nothing recorded" chips exist to make catching up easy, not to nag you.